ThinkHub

Slide 1

Introducing LoadCast

Teaching a regression model to see tomorrow's electricity demand using yesterday's patterns.



What is LoadCast?

An intelligent platform to forecast **electricity demand** using machine learning.

What does the name mean?

LoadCast =
'Load' (Electricity Demand) +
'Cast' (Forecast)

Project Objective

Build a reliable and interpretable forecasting model for **Indian power grid.**

Predicting Tomorrow's Electricity Demand

Using historical patterns to forecast **next-day demand.**

FORECAST • ANALYSE • POWER THE FUTURE

02

The Problem

Electricity demand is dynamic, and grid planners must estimate it in advance to keep the power system reliable, cost-efficient, and balanced.



Why Forecasting Matters?

- Helps schedule power generation in advance.
- **Reduces** operational **cost** and energy waste.
- Ensures grid **stability** and prevents blackouts.

Challenges for Grid Planners

- Demand changes with day, week, season and events.
- Over-forecasting wastes resources and money.
- **Under-forecasting** can lead to supply shortages and outages.

Why Historical Patterns Help?



- Electricity demand shows **repeatable patterns** (daily, weekly, and seasonal).
- **Past consumption** contains **valuable signals** about future demand.
- **Time-series features** (like lags and rolling averages) help the model learn these patterns.

LoadCast

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03

The Solution

Use historical consumption patterns, engineer time-series features, and train a Ridge Regression model to predict tomorrow's electricity demand.

Historical Electricity Data

State-wise daily consumption (MU) from POSOCO (2019 – 2020)

Lag & Rolling Features

Capture time dependencies using lag-1, lag-7 and rolling 7-day average

Ridge Regression

A regularized regression model (L2) that learns the relationship and avoids overfitting

Next-Day Prediction

Estimate tomorrow's electricity demand for Rajasthan

Feature Engineering (Input to Model)

y_{t-1} (Lag-1)	y_{t-7} (Lag-7)	$\bar{y}_7$ (Rolling Mean)	Day of Week	Month	Weekend (0/1)

Ridge Regression (L2)

$$\min \quad \|y - X\beta\|^2 + \alpha\|\beta\|^2$$

- Learns from historical patterns
- L2 regularization controls overfitting
- Robust and interpretable

Next-Day Forecast

205.5 MU

↗ +2.4% (vs. yesterday)

End-to-End Pipeline

Historical Data → Feature Engineering → Temporal Train/Test Split → Ridge Regression → Evaluation (MAPE, RMSE) → **Next-Day Demand Forecast**

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04

Technology Behind LoadCast

A modern full-stack setup that combines data science, machine learning, and an interactive web experience.

Python & Pandas

Data Processing

- Data cleaning and preprocessing
- Time-series feature engineering (lag, rolling, calendar)
- Analysis and model training pipeline

python | pandas

Scikit-learn & Ridge Regression

Machine Learning

- Ridge Regression (L2 regularization) for forecasting
- Chronological train/test split (no data leakage)
- Evaluation with RMSE, MAE, and MAPE

scikit-learn | Ridge (L2)

Next.js & TypeScript

Web Application

- Modern, fast and scalable web app (Next.js)
- Type-safe and maintainable code (TypeScript)
- Server-side APIs for forecasts and analytics

NEXT.JS | TS TypeScript

Recharts & Vercel

Visualization & Deployment

- Interactive and responsive charts (Recharts)
- Rich energy analytics and trends
- Deployed on Vercel for a fast, reliable, and global experience

Recharts | Vercel

End-to-End Tech Stack

Data (POSOCO) → ML Pipeline (Python + scikit-learn) → Web App (Next.js + TypeScript) → Visualizations (Recharts) → Deployment (Vercel)